

GEORGE PAULY

Fullstack / Backend Software Engineer

georgepauly1995@gmail.com | +91 8884562950 | github.com/GeorgePauly95 | linkedin.com/in/george-pauly

PROFESSIONAL SUMMARY

Backend Software Engineer with over four and a half years of experience designing high-performance Python web services, real-time data pipelines, and scalable AI integrations. Combines deep expertise in custom data engineering and complex relational databases to architect secure, data-intensive systems with a rigorous focus on structural integrity and system reliability.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

Backend: FastAPI, Django, Flask, RESTful APIs, WebSockets, async / asyncio, Pydantic

Data: PostgreSQL, SQLAlchemy (ORM), Alembic, pgvector, Redis, MongoDB, OpenRefine

AI & Audio: OpenRouter, WhisperX, Pyannote, Speech-to-Text

Auth: OAuth2 / Google Sign-In, JWT

Infra & Tooling: Docker, Render, Git, uv, pip, pytest, CI/CD

Frontend: React, HTML, CSS

WORK EXPERIENCE

Geekwerke Software | Bangalore

Feb 2025 – Present

Software Engineer

AI-Powered Support Chat Widget

- Owned the backend messaging architecture for a live React-based support widget, engineering a low-latency FastAPI WebSocket layer supporting thousands of concurrent user sessions.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline utilizing pgvector and OpenRouter to deliver context-aware AI draft responses, reducing average ticket resolution time by ~50%.
- Integrated an automated speech-to-text transcription pipeline to process voice queries directly into the text workflow.
- Maintained persistent user authentication across WebSocket connections by configuring Redis for rapid cookie validation and session state management.

GPS Data Tracking Pipeline

- Developed an asynchronous FastAPI tracking service that streamed live surveyor GPS data into PostgreSQL, reliably processing 50,000+ daily location pings from thousands of active daily users.
- Protected sensitive location streams by enforcing JWT validation and AES-encrypted session tokens on top of Google OAuth2 authentication.
- Managed database schema updates with Alembic and independently deployed the layered service to Render.

Cloutrack Software Labs | Bangalore

Feb 2022 – Jan 2025

Lead Data Analyst

Dec 2023 – Jan 2025

- Directed end-to-end data delivery architectures for over 15 enterprise clients, utilizing Python and SQL to automate dashboard generation and decrease manual reporting turnaround time by 40%.
- Managed data extraction and processing strategies, ensuring 99%+ data integrity and structural accuracy across all high-visibility client deliverables.

Data Analyst

June 2022 – Nov 2023

- Engineered and optimized custom Python web scraping and cleaning pipelines, successfully extracting and processing over 1 million unstructured data points monthly with high fault tolerance.
- Owned the end-to-end quality of programmatic data deliverables, coordinating directly with stakeholders to deliver structured datasets consistently ahead of schedule.

Data Intern

Feb 2022 – May 2022

- Developed Python scripts and utilized OpenRefine to extract, clean, and standardize large datasets containing 100,000+ records, reducing downstream analytics preparation time by 25%.

EDUCATION

- Master of Science (M.Sc.) in Mathematics** | Mysore University
- Bachelor of Science (B.Sc.) in Mathematics, Physics, and Electronics** | Christ University